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Collaborative innovations in diabetes self-care for individuals with type 2 diabetes and schizophrenia: A Participatory Design study developing a diagnosis-specific educational manual

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ABSTRACT

Background and aims: Individuals with schizophrenia are at high risk of developing type 2 diabetes. This study aimed to develop a tailored solution to address their complex diabetes care needs, based on insights from patients and healthcare professionals, to enhance self-care.
Methods: Using a Participatory Design approach, we conducted three workshops and an evaluation, which included focus groups, interviews, and written feedback. Patients, healthcare professionals, and stakeholders actively participated in all stages of the process between May 2022 and December 2023. Iterative processes ensured comprehensive input in idea generation and concept development. Data analysis followed the steps of planning, acting, observing, and reflecting. The study is reported using SRQR framework.
Results: Participants highlighted challenges such as navigating a fragmented healthcare system, undertreatment, and stigma. In response, a tailored educational manual for voluntary mentors was developed. This two-day training program equips mentors to support individuals with type 2 diabetes and schizophrenia, fostering collaboration and bridging the gap between psychiatric and somatic care.
Conclusions: A co-designed approach may enhance diabetes self-care and improve coordination between healthcare sectors.

1. Introduction

Extensive documentation has described the correlation between type 2 diabetes (T2D) and schizophrenia, revealing a concerning reduction in life expectancy of up to 20 years for individuals with schizophrenia compared to the general population [1–3]. This premature mortality is partly attributed to the early onset of schizophrenia, which elevates the risk for the development of comorbid conditions such as diabetes [1,4, 5]. Individuals with severe mental illness (SMI), including schizophrenia, often face inadequate treatment and underdiagnosis of medical conditions like diabetes [6–9]. These individuals often miss essential health checks and receive limited diabetes education, which hinders effective diabetes self-care [10].

Qualitative studies highlight the challenges of managing both conditions, including difficulties distinguishing symptoms, maintaining self-care routines, and prioritizing diabetes self-management during mentally stressful periods (3, 8, 11 [11], 12, First author13). Establishing structured care routines is often challenging, leading to feelings of self-blame and frustrations (First author13, 14 [12]). Despite these challenges, there remains a lack of holistic approaches to caring for individuals with co-occurring T2D and schizophrenia (8, First author13, First author15).

Research has explored the potential benefits of integrated care, demonstrating that incorporating patient perspectives enhances the effectiveness of combined consultations. Studies indicate that when psychiatric and diabetes care are delivered in an integrated manner, patients experience improved health outcomes, better adherence to treatment, and a greater sense of support in managing both conditions

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[13–17]. Conversely, independent consultations by psychiatric and diabetes specialists often result in fragmented care, as they focus primarily on their respective areas of expertise, neglecting the interconnected nature of mental and physical health challenges [17].

Despite the promise of integrated care, its implementation is hindered by systemic barriers, including structural inefficiencies, insufficient communication between healthcare professionals (HCPs), and an unequal prioritization of mental and physical health needs [8,9,18,19,20,16,21–26]. These obstacles contribute to the ongoing fragmentation of care, reinforcing the need for innovative approaches that bridge the gap between psychiatric and somatic healthcare services.

Despite growing recognition of the need for integrated care, research on developing patient-centered interventions for individuals in vulnerable positions remains limited [27–29]. Co-design and co-creation approaches are rarely applied, often due to assumptions about patient competencies and systemic barriers within healthcare settings. This increases the risk of developing interventions that overlook patient voices and real-world needs.

Thus, this study aims to explore how a co-designed intervention can best support diabetes self-care and interdisciplinary collaboration, ultimately improving diabetes self-care for individuals with schizophrenia and T2D. We hypothesize that applying co-design approaches will enhance user involvement in healthcare decision-making, leading to more effective and sustainable care solutions.

2. Methods

2.1. Design

The overall framework for this study was Participatory Design (PD), a user-driven approach that emphasizes collaboration and co-creation with the individuals experiencing the challenges at hand [30,31]. Building on previous research exploring the challenges and needs in diabetes care for individuals with schizophrenia and T2D (First author13), this study aimed to design and develop a solution through workshops involving patients, HCPs, and researchers. The solution was subsequently evaluated through focus group interviews, individual interviews, and written feedback.

PD is defined as "a process of investigating, understanding, reflecting upon, establishing, developing and supporting mutual learning between multiple participants" [31]. It fosters active participation throughout the process, making it a relevant methodology for addressing complex healthcare challenges by involving those directly affected. As an extension of action research, PD focuses on solving practical problems while bridging the gap between theory, research and practice [31]. The iterative PD process typically involves three overlapping **phases**: 1) Identification of needs, 2) Design and development, and 3) Testing in clinical practice and evaluation [30], with iterations occurring within each stage. This study focused on phase 2.

A cornerstone of PD is genuine participation, ensuring that patients have a "voice" throughout the research process, which is essential for a successful implementation of the developed solution [30,32,33]. From the early planning stages, individuals with schizophrenia and T2D, alongside HCPs, were actively involved to identify unique challenges, generate ideas, and co-create solutions to improve diabetes care. The active participation of these key stakeholders was crucial for ensuring that the final solution directly addressed their needs and challenges. Fig. 1 provides an overview of the study's design and development processes.

2.2. Setting and participants

Three collaborative workshops were conducted in two different settings: Workshop 1 and Workshop 3 took place at [blinded], an outpatient clinic specializing in the diagnosis and treatment of mental health disorders, familiar to individuals with schizophrenia and T2D. Workshop 2 was held at a neutral university location [blinded] to provide a more comfortable, non-clinical environment. The choice of settings was guided by patient preferences and the need to balance familiarity with neutrality, ensuring that participants felt at ease throughout the process.

Additionally, the evaluation methods were conducted within the setting of two online focus groups, one individual online interview with a diabetes consultant, and written feedback from one patient, all contributing to comprehensive input for the final design. The first focus group included one HCP from somatic care and one from psychiatric care, while the second focus group consisted of four representatives from a non-governmental organization (NGO) focused on health equity. All sessions were recorded to capture detailed insights, which were subsequently incorporated into the revised version of the solution.

While demographic details such as ethnicity, gender, and age were not formally collected, the groups included a diverse range of patients and professionals. Table 1 outlines the specific titles and roles of participants across the various phases of the study.

As all participants were fluent in the primary language of communication, interpretation services were not needed. The first author, a

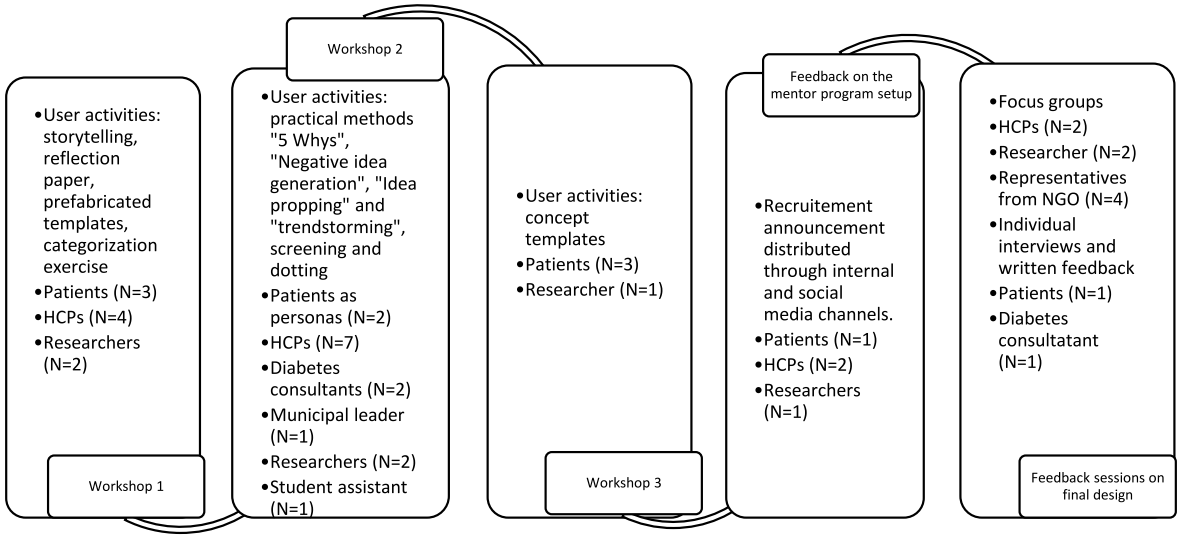


Fig. 1. An overview of the study's design and development processes.

Table 1
Participant involvement in workshops and feedback sessions.

Participants	Workshop 1	Workshop 2	Workshop 3 –Mock-up	Feedback on the advertisement for volunteers	Feedback sessions on final design
Patient 1	X				
Patient 2	X				
Patient 3	X		X		
Patient 4			X		
Patient 5			X		
Patient 6				X	
Patient 7					X
Nurse Psychiatry	X	X		X	X
Occupational therapist psychiatry	X	X			
Medical doctor psychiatry	X	X			
Nurse diabetes	X	X			X
Consultant diabetes		X			
Consultant diabetes		X			
Consultant diabetes				X	X
Medical doctor		X			
General practitioner		X			
Nurse municipal		X			
Leader municipal		X			
NGO, person 1					X
NGO, person 2					X
NGO, person 3					X
NGO, person 4					X
Researcher	X	X	X	X	X
Researcher	X	X			X
Student assistant		X			

registered nurse and Ph.D. candidate specializing in diabetes, facilitated the study, which was conducted between May 2022 and December 2023.

Recruitment primarily included individuals who had participated in phase 1 of the study, along with additional HCPs with relevant clinical expertise. The first author invited participants via email, ensuring a broad representation of both patient perspectives and professional experiences.

Recognizing the challenges of involving individuals with both conditions in every phase, we utilized personas in Workshop 2, developed from narrative findings in phase 1. These personas played a pivotal role in ensuring that the perspectives of individuals with schizophrenia and T2D were consistently represented, even when direct participation was not feasible due to safety concerns raised by HCPs. The use of personas allowed discussions and decisions to remain grounded in patient experiences, supporting the co-creation process.

A fundamental principle of PD is to equalize power dynamics and actively involve participants in decision-making, particularly those who are often marginalized in healthcare settings [28]. While we aimed for full participation of individuals with T2D and schizophrenia throughout the study, the safety concerns in Workshop 2 necessitated the use of personas. This adjustment enabled continued representation of patient voices in meaningful ways while addressing practical and ethical concerns of direct involvement [19,34–36].

2.3. Data collection

Data collection followed an iterative process of planning, acting, observing, and reflecting [30,37]. Workshops, focus groups, and interviews were designed to facilitate a flexible and responsive approach that systematically integrated the design and development process [38].

Workshops originated from innovative group approaches to problem-solving through brainstorming, reflection, and exploration of specific issues [39]. As a research methodology, workshops are intended to generate actionable data to guide future design processes and address organizational challenges [40]. Prior to each workshop, careful planning took place, including detailed descriptions of purpose, desired outcomes, and processes [41–43]. The iterative nature of the workshops allowed for flexibility, adjusting based on the experiences and feedback obtained from previous sessions.

To ensure active user involvement, various tools and techniques were employed throughout the workshops, grouped into three categories of activities: “telling, making, and enacting.” “Telling” involved participants sharing their narratives, knowledge, and wishes for the future. “Making” focused on creating artifacts and visual representations of ideas, enabling participants to concretize future scenarios. “Enacting” encouraged ongoing evaluation and refinement through visualization and discussion of work-in-progress concepts. This structured approach ensured that each workshop progressed from critical reflection to envisioning and, ultimately, testing the feasibility of the proposed solutions.

The research team, consisting of the primary author and her supervisory group, shared reflections after each workshop, promoting collective learning and enhancing the iterative nature of the process. The data generated from the workshops included notes, post-its, posters, photos, and audio recordings, ensuring a rich data set to support the study’s findings. Reporting was guided by the Standard for Reporting Qualitative Studies (SRQR) [44].

2.3.1. *Future Workshops*

Workshops 1 and 2 were designed to present findings from the earlier needs assessment phase (phase 1 of PD) [19] and to facilitate discussions on improving diabetes care for individuals with T2D and schizophrenia. The Future Workshop methodology, comprising three distinct phases—1) a critique phase, 2) a vision phase, and 3) a reality phase, was used to guide the process.

2.3.2. *Workshop 1*

The first workshop took place in May 2022 at the outpatient psychiatric clinic. During this 2-h session, participants generated ideas and suggestions for enhancing diabetes care [45].

The workshop began with an icebreaker, creating a relaxed atmosphere for open discussion. Following this, findings from the ethnographic field studies conducted in phase 1 of PD [19] were presented in a storytelling format, developed by a professional storytelling firm. This method engaged participants emotionally and encouraged deeper reflection [46].

Participants were then divided into smaller groups to critically assess current diabetes care, documenting their critiques on pre-made templates. Ideas were noted on post-its and placed on posters, facilitating

collaborative idea generation. During the vision phase, participants categorized the critique points using a color-coded system (green, blue, and red), symbolizing ski slopes, to prioritize challenges and envision future treatment options. The final reality phase focused on how these visions could be realized, incorporating both small and significant adjustments to current practices.

2.3.3. Workshop 2

The second workshop, held in September 2022 at a university setting, built upon the findings from Workshop 1 to refine ideas for optimizing diabetes care. This 5-h Future Workshop began with a critique phase, where participants reviewed findings from both phase 1 of PD [19] and Workshop 1 to reestablish the key challenges.

The "5 Whys" tool, a method of repeatedly asking "why" to uncover the root cause of challenges, was employed in smaller groups to explore why adhering to diabetes care can be particularly challenging for individuals with schizophrenia. Following the critique phase, the vision phase used the "Negative Idea Generation" tool to transform critical points into positive, actionable ideas [41,47]. These ideas were then prioritized using a dot-voting system. In the reality phase, participants worked at three stations using tangible elements to bring their ideas to life (Fig. 2) [41,46,47]. A final wrap-up session ranked the ideas based on their feasibility, using a "Potential Pyramid" to categorize them into top, second-best, and least-prioritized concepts [47]. Fig. 2 shows participants working at stations with tangible elements to facilitate the design process.

2.3.4. Workshop 3

As a continuation of the iterative design process, Workshop 3 aimed to reach consensus on a patient-validated concept for the future intervention. This workshop involved individuals with schizophrenia and T2D in one-to-one sessions with the first author and played a crucial role in validating ideas generated in earlier sessions. Visual templates were created based on the five most significant ideas from the previous workshops [1]: a shared communication platform [2], future healthcare offerings [3], a dream scenario [4], education of HCPs, and [5] communication with HCPs. Fig. 3 illustrates the design of the templates created based on the five most significant ideas from the previous workshops.

Each concept was discussed in detail using the "Idea Improvement" tool [48], a method designed to identify both strengths and areas for improvement by systematically analyzing each idea. Participants engaged actively in refining these ideas, ensuring that patient perspectives were central to the development process. For example, participants articulated how a shared communication platform could streamline care, while the "dream scenario" encouraged discussion on the impact of more personalized support, such as mentorship. Additionally, the "education of HCPs'" template was used to gather feedback on how HCPs could improve their competencies, particularly regarding patient involvement in their own care.

Finally, participants ranked the five concepts based on what they believed should be prioritized in their interactions with the healthcare system, ensuring that the patient voice remained integral throughout the design process.

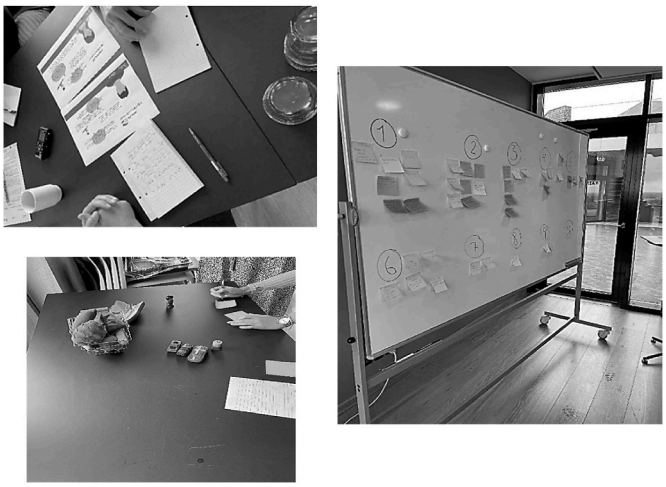


Fig. 2. Participants working with personas, tangible elements and posters.

2.3.5. Development of a mentor/mentee setup

To translate the results (section 3) from the three workshops into practice, the research group initially developed a mentor/mentee program. This was presented as a low-fidelity mock-up [49], designed to engage individuals with experience in T2D alone, as well as those managing both T2D and schizophrenia. The mentorship program aimed to help patients navigate various aspects of their lives with diabetes while managing co-occurring schizophrenia. By involving individuals with diabetes experience as mentors, we sought to provide participants with tailored knowledge, practical assistance, and emotional support to address their specific needs.

Based on feedback, the mentor/mentee program was adapted into a social media campaign and distributed across various digital platforms, patient association networks, larger patient organizations, and in the respective clinics of both diabetes and psychiatry.

However, recruiting voluntary individuals with experience in T2D alone proved challenging. Many volunteers with T2D were over the age of 60, leading to a mismatch with the target demographic, and there was also a lack of willingness to participate. These challenges prompted the research team to reassess the recruitment strategy and explore alternative approaches.

Through a thorough literature review of existing mentor/mentee programs in Denmark within the diabetes field, it became clear that implementing such a program would require a stronger foundation, including the development of diagnosis-specific services. As a result, a diagnosis-specific educational manual for future volunteer mentors was developed. The manual was informed by the comprehensive literature review, alongside insights from earlier workshops and participant feedback. Instead of involving individuals with T2D as mentors, the manual focuses on having the mentoring role carried out by volunteers from NGOs with a background in health-related education.

2.3.6. Feedback on the diagnosis-specific educational manual

Following the development of the diagnosis-specific educational manual, a comprehensive feedback process was carried out to ensure it addressed participants' needs. This process included two online focus groups, one individual interview, and written feedback from one patient. Participants were provided with focused questions in advance, such as: 'Is the content clear and understandable?', 'Are there areas that need further elaboration?', and 'What improvements could enhance the effectiveness of the material?'. The feedback gathered provided valuable insights into the manual's structure, clarity, and potential areas for improvement. These suggestions were then used to refine the manual iteratively, ensuring that it effectively conveyed its intended message.

2.4. Data analysis

After each workshop, all collected materials were transcribed and supplemented with visual documentation of the processes in action. This ongoing reflection and documentation ensured transparency and allowed the research team to continuously refine the study's direction, aligning with the iterative nature of the design process. The feedback from participants was integrated into the analysis, helping to adjust the educational manual and the overall approach [50].

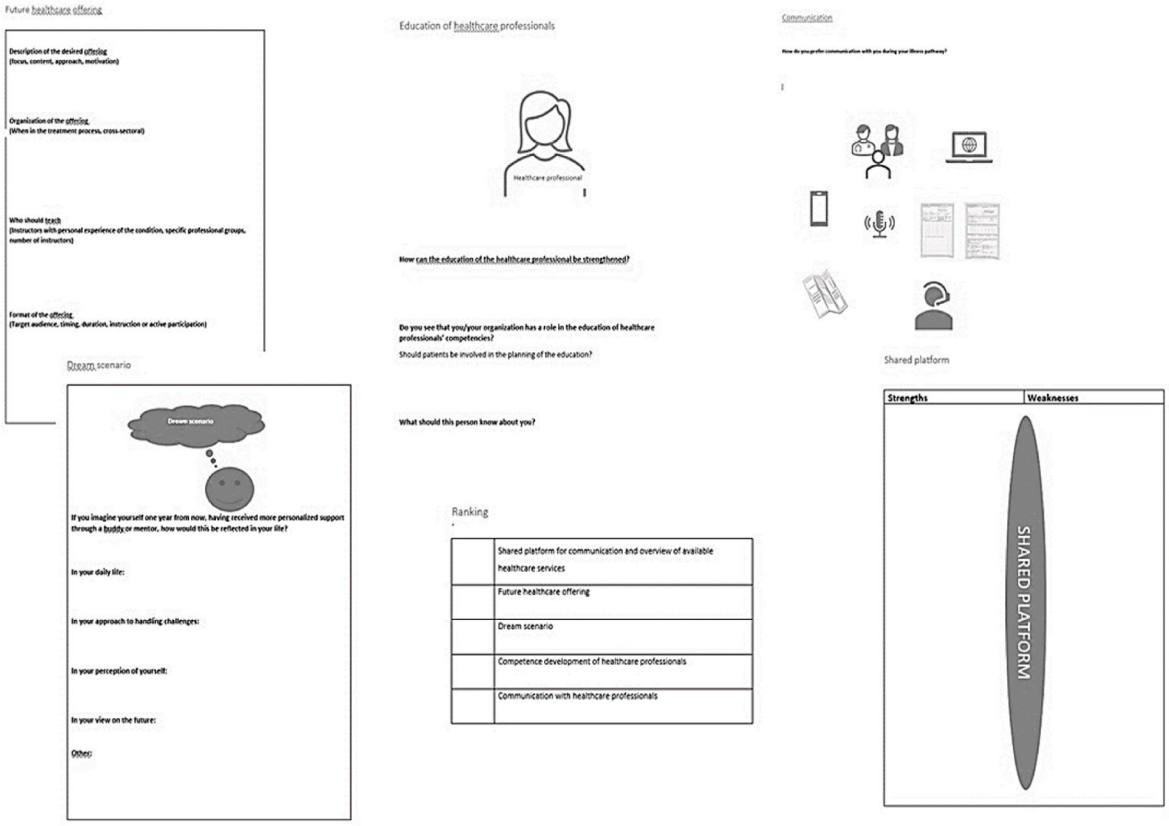


Fig. 3. The five concepts templates from workshop 3.

2.5. Ethical considerations

Ethical approval was obtained from the Danish Data Protection Agency (journal number: 19/45675). Participants were given oral and written information about the study aim, confidentiality, and the voluntary nature of participation before signing consent forms [51].

3. Results

Building upon the iterative workshops and continuous feedback from patients, HCPs, and volunteers, the workshops provided several suggestions for an intervention supporting self-care in individuals with T2D and schizophrenia.

3.1. A patient-centered approach

The patient-centered approach, emphasized in earlier workshops, was reinforced in future sessions with specific suggestions, including the need for more individualized treatment, peer-to-peer support, and a stronger focus on reducing stigmatization and enhancing patient empowerment. The storytelling exercise conducted in Workshop 1 evoked various emotional reactions, including feelings of both stigmatization and identification, highlighting the complex and diverse experiences of patients. One patient noted:

> "Just because we have schizophrenia doesn't mean we have low IQ" (Patient).

Further, participants suggested that support inspired by the peer-to-peer model, ideally provided by an individual with lived experience of T2D, could be beneficial. However, implementing such support was recognized as challenging due to the variability in services offered across different regions and the inconsistent continuity of care. Both patients and HCPs acknowledged the disparities in available services, which were often difficult to navigate due to fragmented information spread across digital platforms.

HCPs emphasized the need for more visibility and equal access to healthcare services, ensuring uniformity in treatment approaches. One HCP suggested:

> "Utilize peers. Everyone should have the opportunity to have a peer. We need to focus more on somatic comorbidities and the use of peers in somatics. We don't have that" (post-it note, HCP).

The solutions proposed in Workshop 2 included interdisciplinary approaches, platforms for mutual communication, and the use of technological tools to help provide better visibility of healthcare services. These suggestions were aimed at addressing the challenges of navigating and accessing available services, while also ensuring offerings were tailored to individual patient needs.

The ideas emerging from these discussions were summarized into five final concepts, which were reviewed and refined during Workshop 3, forming the foundation for the next phase of development.

3.2. Mentor support with diabetes experience

In Workshop 3, patients were presented with five concepts based on earlier feedback, with a focus on improving the visibility of healthcare services through a shared digital platform. Among these concepts, the shared platform was the most preferred by the HCPs, as it offered a comprehensive solution for connecting patients with the right services. However, while some patients supported the idea of the platform, others raised concerns about its accessibility, particularly for those who struggle with using digital solutions. One patient pointed out:

> "It's challenging to discern what might be relevant or where within the platform one should locate them (ed. Services). It's difficult to determine which subgroups one belongs to, despite attempts to

categorize offerings, and where to begin if aware of a participating service. Needs vary on an individual basis" (Patient).

Additionally, many patients emphasized the value of in-person appointments and the necessity for daily personal support to manage their diabetes. As one patient explained:

"Even when well-treated for schizophrenia, I may still need assistance with my diabetes during periods of heightened symptoms in schizophrenia" (Patient).

In response to this experience, the research group recognized the potential of mentor support as a way to help individuals navigate diabetes management across different life domains. This led to a deeper exploration of the concept, with particular attention to the challenges patients face in accessing existing peer support services in psychiatry, which they viewed as valuable but often difficult to engage with when they were well-managed for schizophrenia.

Feedback on the first version of the mentor program clarified the role of volunteer mentors. Two HCPs and one patient reviewed the setup, providing valuable insights into the need for clear communication regarding the mentor's function. As one HCP emphasized:

"It's important that the description includes what the volunteer mentor can offer and is valued for, and that clarity is created around the function the mentor is expected to fulfill in the process" (HCP)

This feedback prompted revisions that refined the mentor's role and responsibilities, ensuring that both the program's purpose and the volunteer's contributions were clear and well-defined before the recruitment announcement was distributed.

3.3. Developing a diagnosis-specific educational manual for future voluntary mentors

In response to the challenges faced in recruiting mentors, the next critical step was the development of an educational manual for future volunteer mentors, particularly those from NGOs or similar volunteer-based organizations. The manual was created in collaboration with a design and branding agency, the manual incorporated insights from both literature on mentor programs [52–54] and participant experiences. Its aim was to facilitate partnerships between voluntary and public organizations through network-based collaboration.

Designed specifically for mentors with a clinical background, such as those from health-related NGOs, the manual serves as both a 2-day educational program and a reference guide, supporting mentors as they work with individuals managing T2D and schizophrenia. It includes sections on understanding the two conditions, mentor training, communication skills, and collaboration with the healthcare system. The content is designed to help mentors support patients in managing their diabetes while navigating the complexities of schizophrenia. Fig. 4 visualizes the design of the manual, with a contents list (in Danish).

The manual underwent revisions based on ongoing feedback, with particular focus for its clarity, adaptability, and organizational sustainability. Participants emphasized the need for strong partnerships and suggested a few key improvements, such as increased diversity in representation, simplified content, and enhanced communication methods. One participant noted:

"It could be beneficial to have patient representatives as educators. It (ed. the manual) gets very clinical orientated, so one tends to overlook the life story within it" (Representative from the NGO).

Additionally, the feedback highlighted the need for regular supervision and psychological support for mentors. Flexibility was deemed essential to accommodate the diverse needs of both mentors and mentees. This iterative feedback process led to significant revisions, ensuring the manual better met participants' needs and expectations.

Ultimately, the manual was designed to be a sustainable and practical resource for mentors, ensuring they are equipped to foster meaningful relationships with mentees and collaborate effectively with the healthcare system. Link to educational manual in Danish (First author55 [55]).

4. Discussion

The iterative workshop process generated practical solutions, such as peer-inspired mentorship programs and clearer communication pathways in healthcare. These initiatives directly address key challenges, including stigma, fragmented support, and unclear roles, emphasizing the need for a patient-centered approach that considers the complex care needs of individuals with T2D and schizophrenia.

A key theme that emerged during the workshops was the stigma surrounding both mental health and diabetes. This stigma affects self-



Fig. 4. Educational manual and table of content (First author55 [55]).

esteem and discourages engagement with treatment and support systems. Research indicates that social and peer support can positively impact both mental and physical health outcomes, particularly self-management, among individuals with diabetes and SMI [56–60].

Consistent with findings by Blixen et al. [60], our study highlighted the potential of peer mentorship in managing chronic conditions. Workshop participants emphasized the importance of peer mentors with lived experience of diabetes, reinforcing previous findings that peer-based programs enhance self-management and provide essential emotional support [61]. However, implementing such programs for individuals with dual diagnoses presents challenges related to stigma and organizational shortcomings.

The stigma associated with a dual diagnosis of T2D and schizophrenia can lead to social isolation and reluctance to seek support. Peer mentorship emerged as a promising strategy to overcome these barriers by offering empathetic, tailored guidance. To support this, the educational manual was developed to equip future mentors with practical tools, including communication techniques and strategies to build trust, helping to reduce the shame and hesitation that many patients experience in healthcare interactions. As Levine and Nickel highlight, personalized and empathetic support is instrumental in breaking down stigma and fostering a more supportive environment for self-care [62, 63].

Workshop discussions also underscored the fragmentation of healthcare services, particularly the disconnect between mental health and diabetes care. Patients often experience confusion and frustration when navigating disjointed services, while HCPs report difficulties in coordinating care across disciplines. The educational manual for mentors was designed to address this issue by defining mentor roles and establishing clear guidelines for collaboration with HCPs. This structured approach bridges the gap between psychiatric and somatic care, ensuring that mentors can guide patients effectively through the healthcare system.

Both patients and HCPs expressed the importance of clear role definitions for mentors to enhance program adherence and effectiveness. A review by Coxen et al. examined the effectiveness of diabetes self-management programs for the general diabetes population and found that individuals with SMI are often excluded from such trials. This exclusion results in a lack of evidence regarding suitable interventions for managing diabetes in this population. The study highlights the need to adapt existing programs or develop new interventions tailored to individuals with SMI [64]. Similarly, Danish health authorities advocate for more diagnosis-specific initiatives to address the distinct needs of individuals with complex conditions [65,66]. Findings from both the literature and our study indicate significant opportunities for integrating mentor support into future interventions for individuals with T2D and schizophrenia.

The adoption of a PD approach in this study represents a significant step in addressing the complex needs of individuals with T2D and schizophrenia. By actively involving patients, HCPs, and researchers in the design process, the study embodies a co-creation model that prioritizes lived experiences. This participatory approach not only empowers patients but also facilitates a deeper understanding of the interplay between physical and mental health [67,68]. However, existing literature notes a lack of robust evaluations of interventions derived from co-approaches, highlighting the need for further research [69].The iterative development of the educational manual was informed by continuous feedback from patients and healthcare professionals, ensuring that it aligns with real-world challenges. More than just a training tool, the manual represents a collaborative effort to empower mentors and address the specific needs of their patients.

Given these insights, the creation of a diagnosis-specific educational manual for mentors fills a critical gap in both research and practice [22]. By equipping mentors with the necessary knowledge and skills, the manual supports a more integrated approach to care, ultimately improving outcomes for this vulnerable population.

This study lays the groundwork for future research and initiatives aimed at integrating mentor support into patient-centered care for individuals with T2D and schizophrenia. Addressing both structural and stigma-related barriers, these findings contribute to a growing body of knowledge advocating for more inclusive and effective healthcare practices, ensuring that individuals with dual diagnoses receive the comprehensive support they need.

4.1. Limitations

Further research is needed to validate the effectiveness and generalizability of these co-approaches across different regions and healthcare systems. Recruitment for the mentorship program faced significant challenges, particularly in attracting individuals with T2D experience and collaboration with organizations, highlighting the need for stronger organizational support.

Involving patients as key users in workshops required extensive time and planning, which may impact engagement levels. Additionally, the limited sample size and diversity may restrict the generalizability of findings. Nonetheless, it is essential to integrate individuals in vulnerable positions into PD processes to understand their specific needs and enable them to contribute to the development of effective support initiatives.

5. Conclusion

This user-driven study and iterative workshops unveiled critical insights into improving diabetes care for people with schizophrenia and T2D. By adopting a PD approach, a nuanced understanding of patient experiences was gained, emphasizing the significance of inclusive, individualized, and diagnosis-specific initiatives. The findings align with existing literature, reinforcing the need for continuous evaluation and adaptation to handle the multifaceted challenges in integrated care.

The identified challenges in implementing a diagnosis-specific mentorship program present opportunities for innovation. Moving forward, a critical next step is piloting the educational manual in clinical settings to assess its effectiveness, gathering feedback from mentors and healthcare professionals to refine the program and optimize its impact on patient outcomes. By assessing its impact on mentor preparedness, patient engagement, and self-management outcomes, this initiative aims to bridge existing gaps in support and supplement current medical treatments for individuals with T2D and schizophrenia. Furthermore, refining mentorship structures based on real-world feedback will be key to ensuring sustainability and effectiveness.

By addressing both structural and stigma-related barriers, this study lays the foundation for sustainable, patient-centered interventions that can transform care for individuals living with T2D and schizophrenia.

5.1. Relevance for clinical practice

The study highlights the importance of a patient-centered approach and the need to reduce stigma in clinical practice for individuals with both T2D and schizophrenia. It underscores the necessity for individualized interventions and diagnosis-specific support, such as mentorship programs. To enhance implementation, HCPs could benefit from structured training on integrating peer mentors into multidisciplinary care teams, ensuring that mentorship initiatives are aligned with existing treatment pathways. Moreover, awareness campaigns within healthcare settings may help reduce implicit stereotyping and encourage a more holistic approach to dual-diagnosis care. By integrating the voices and needs of individuals in vulnerable positions into the design process, these insights could lead to more effective and tailored care strategies, ultimately bridging the gap between mental health and diabetes management.

Contributions each author

Tanja Juhl Mikkelsen: Conceptualization, Data curation, Formal analysis, Study design, Methodology, Project administration, writing original draft, writing manuscript and editing. **Dorte Moeller Jensen and Elsebeth Stenager:** Conceptualization, Study Design, writing manuscript and editing. **Mette Juel Rothmann:** Conceptualization, Formal analysis, Methodology, Study Design, writing manuscript and editing.

Ethics approval statement

Ethical approval was obtained from the Danish Data Protection Agency (journal number: 19/45675.

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Declaration of competing interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.dsx.2025.103220.

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